Finite Groups and Their Representations — Exercise Sheet 4

Simple, solvable and nilpotent groups

Variant: Questions only

Exercise 1 (When an alternating-group class splits). Let $n \geq 2$ and $\sigma \in A_n$ have cycle type μ. Prove that its S_n-class splits into two A_n-classes exactly when all parts of μ are odd and pairwise distinct. Include parts equal to 1. Apply the criterion to types $(5,3,1)$, $(3^2,1^3)$ and $(4,2,1^3)$ in A_9, and find the alternating-group class sizes.

Exercise 2 (Prime-order composition factors). Prove that a finite group is solvable if and only if all its composition factors have prime order. First prove that a finite abelian simple group has prime order. Include the trivial group.

Exercise 3 (The dihedral abelianisation). For $n \geq 3$, compute D_n' and D_n/D_n'. Find the derived series and the number of homomorphisms $D_n \to \{1,-1\}$.

Exercise 4 (Simplicity of A_n). Prove that A_n is simple for all $n \geq 5$, extending the A_5 theorem of Lecture 4. Follow this induction.

(i) Prove that three-cycles generate A_n, and that a permutation commuting with all three-cycles is the identity when $n \geq 5$.

(ii) Let $N \trianglelefteq A_n$ be non-trivial. Show that if N contains a non-identity element fixing a point, induction makes $N = A_n$.

(iii) For $n \geq 7$, take $g \in N \setminus \{e\}$ and a three-cycle τ not commuting with it. Show that $[g,\tau]$ moves at most six points. For $n = 6$, treat the even permutations with no fixed points, of types $(3,3)$ and $(4,2)$, by suitable three-cycles.

(iv) Deduce that for $n \geq 5$ the derived series of S_n is $S_n \trianglerighteq A_n \trianglerighteq A_n \trianglerighteq \cdots$.

Exercise 5 (Nilpotency and Sylow subgroups). Prove that a finite group is nilpotent if and only if it is the direct product of its Sylow subgroups, one for each prime dividing its order. Include the trivial group as the empty product.

(i) For nilpotent G and $H < G$, use the least j with $Z_j \not\leq H$ to show $H < N_G(H)$.

(ii) For a Sylow subgroup P, put $K = N_G(P)$. Show that $K < G$ would contradict the first part, using Sylow conjugacy inside K.

(iii) Show that normal Sylow subgroups commute and give the direct product. For the converse compare the upper central series of a direct product with those of its factors.

Exercise 6 (Nilpotent groups of order 1800). Classify the nilpotent groups of order $1800 = 2^3 \cdot 3^2 \cdot 5^2$ by listing the possible Sylow factors. How many isomorphism types occur? How many are abelian? Use the classifications of order eight on Sheet 3 and order p^2 in Lecture 3.

Exercise 7 ⋆. Let G be the rotation group of a regular icosahedron. You may use the following geometric facts: $|G| = 60$; the non-identity rotations are the 15 edge-axis half-turns, 20 face-axis rotations of order 3, and 24 vertex-axis rotations of order 5; conjugacy preserves rotation angle; and G is transitive on vertices, edges and faces. Show $G \cong A_5$. Use the axis counts to find its conjugacy classes, then consider its action on Sylow 2-subgroups. No classification of groups of order 60 may be used.